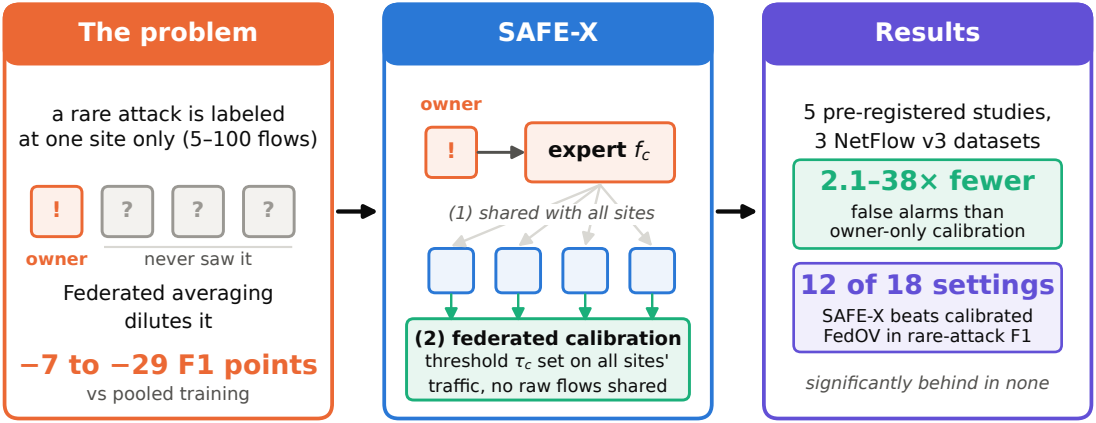
The problem
a rare attack is labeled at one site only (5–100 flows)
!
?
?
?
owner
never saw it
Federated averaging dilutes it
−7 to −29 F1 points
vs pooled training
SAFE-X
owner
!
expert f_c
(1) shared with all sites
(2) federated calibration
threshold τ_c set on all sites' traffic, no raw flows shared
Results
5 pre-registered studies, 3 NetFlow v3 datasets
2.1–38× fewer
false alarms than owner-only calibration
12 of 18 settings
SAFE-X beats calibrated FedOV in rare-attack F1
significantly behind in none